ECE 3710 Lab 4 – Fall 2023

Due Date: Week of October 16 before the beginning of your lab section (two-week lab)

Objectives

The purposes of this lab are to gain experience:

1. Interfacing the microcontroller with a Liquid Crystal Display (LCD)
2. Programming microcontrollers in C
3. Reading and understanding LCD datasheets

Overview

For this lab, you will be creating C functions in a library that controls the LCD included in your lab kit.

Preparation

1. Come with the following:
    a. ECE 3710 Lab Kit
    b. NUCLEO-64 Board
    c. Textbook
2. Read the Midas MD21605G12W3 Datasheet
3. Familiarize yourself with stm32l476xx.h, LCD.h, LCD.c, and main.c that are included in the lab materials.
4. Complete the Pre-Lab requirements and come to lab prepared to show your schematic to the TA.

Pre-Lab

1. There is an LCD in your lab kit. It has 16 holes across the top to facilitate electrical connections. There is also a 16-pin, 0.1 inch spacing header in your lab kit. Carefully

solder the header onto the LCD. There is a soldering iron available for use in the lab room. **Make sure you turn off the soldering iron when you are done using it.**

2. The LCD will be interfaced to the STM32L476 using the 4-bit bus approach shown on page 18 of the LCD datasheet. Draw a schematic diagram of the microcontroller, LCD, and any other components you will need. The LCD runs on 5V power. Make sure the system has a common ground. Turn the backlight on.

3. **Pass your schematic off to the TA.**

Procedure

1. **In this lab, we will use the 16 MHz Internal High Speed Oscillator (HSI) as the clock source.** Look at Chapter 6 of the STM32L4x5 Reference Manual to learn how to manipulate the Clock Control Register (RCC_CR) appropriately.

2. You are required to implement six functions in LCD.c. Develop appropriate pseudocode for each function:

    a. `LCD_Init()` enables GPIO clocks and configures GPIO pins to control the LCD 4-bit bus and the E, RW, and RS lines. It then sends the following commands to the LCD:
        i. 4-bit bus, 2-line display mode, 5x8 dot matrix
        ii. display on, cursor off, cursor blink off
        iii. cursor moves left-to-right, no display scrolling
        iv. clear display
        v. set up CGRAM address to start at 0

    b. `LCD_WriteCom(unsigned char com)` writes the command in `com` to the LCD. Study the timing diagram and reference code provided in the LCD datasheet to assist you in your design.

    c. `LCD_WriteData(unsigned char dat)` writes the data in `dat` to the LCD. Study the timing diagram and reference code provided in the LCD datasheet to assist you in your design.

    d. `LCD_DisplayString(unsigned int line, unsigned char *ptr)` displays text on the LCD, where `line` is the line number (0 or 1) and `ptr` is a pointer to a C-style string (i.e. null-character terminated).

e. `LCD_Clear()` clears the LCD screen.

f. `delay_ms(unsigned int ms)` waits for `ms` milliseconds.

3. **Pass-off** – Demonstrate your working program that successfully prints:

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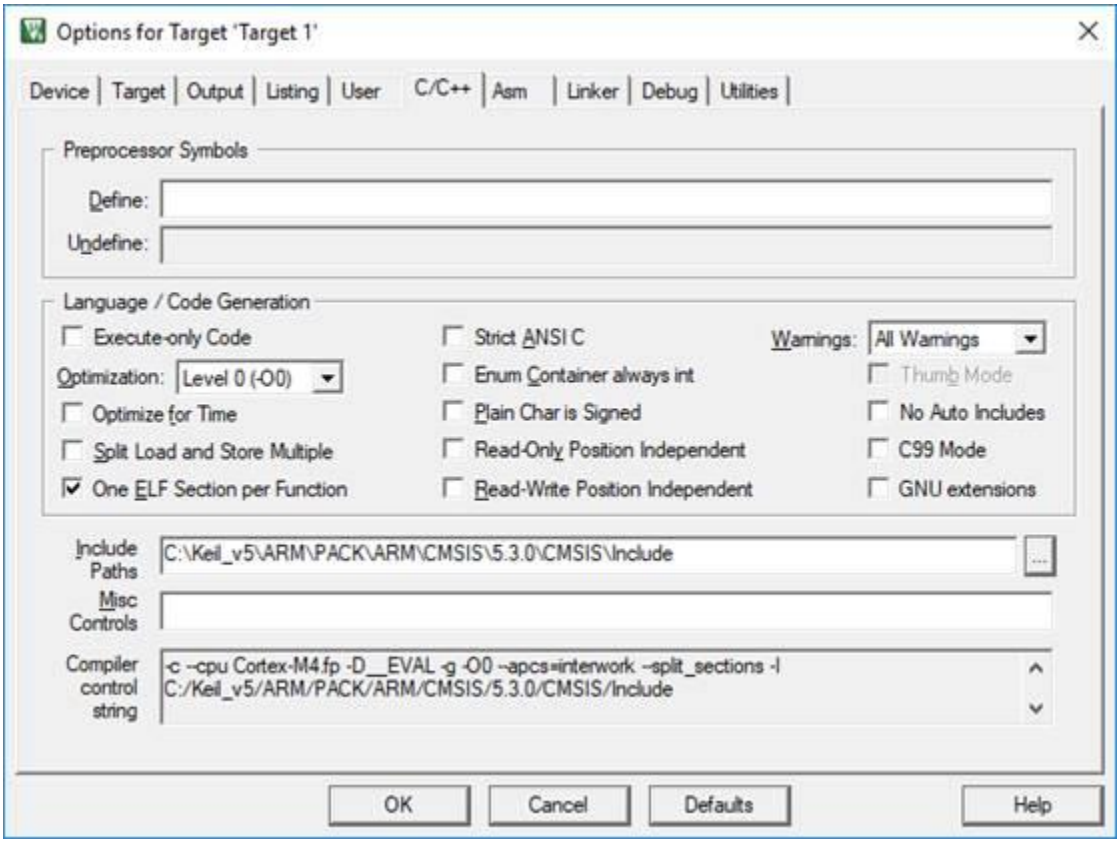
on the LCD to the TA.

Documentation

Prepare your lab report following the same style and rubric that you've followed in previous labs.

Hint

The Keil C compiler needs access to library headers such as core_cm4.h. If you get compiler errors complaining that this (or other similar) files cannot be found, click on **Project -> Options for Target 'Target 1'**. Click on the C/C++ tab and add the include path for these libraries. On my PC, the path looks like this:


Options for Target 'Target 1'
Device
Target
Output
Listing
User
C/C++
Asm
Linker
Debug
Utilities
Preprocessor Symbols
Define:
Undefine:
Language / Code Generation
Execute-only Code
Optimization: Level 0 (-O0)
Optimize for Time
Split Load and Store Multiple
One ELF Section per Function
Strict ANSI C
Enum Container always int
Plain Char is Signed
Read-Only Position Independent
Read-Write Position Independent
Warnings: All Warnings
Thumb Mode
No Auto Includes
C99 Mode
GNU extensions
Include Paths
C:\Keil_v5\ARM\PACK\ARM\CMSIS\5.3.0\CMSIS\Include
Misc Controls
Compiler control string
-c --cpu Cortex-M4.fp -D__EVAL -g -O0 --apcs=interwork --split_sections -I
C:/Keil_v5/ARM/PACK/ARM/CMSIS/5.3.0/CMSIS/Include
OK
Cancel
Defaults
Help